Sharp quartic absorption–scattering asymmetry for bounded-loss reciprocal Maxwell scatterers

A uniform theorem on measurable supports and a finite-sphere realization with an explicit full-vector remainder

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Abstract

For a compact nonmagnetic reciprocal dielectric scatterer in vacuum, let Γ be the absolute difference of absorption cross sections under opposite, time-reversed, real linearly polarized plane waves, divided by their common extinction. Suppose the local susceptibility has an operator-norm cap M and a full tensor loss floor $m > 0$, independently of size and geometry. We prove the exact inequality

$$\Gamma \le \min\left\{1, \frac{2}{3\pi}k^4 R\,|D|\left(m^{-1}+1+\frac{3}{4}k^2a_v^2\right)\left(\frac{M^2}{m}\right)^2\right\}, \qquad a_v = \left(\frac{3\,|D|}{4\pi}\right)^{1/3},$$

for every positive-volume measurable support $D \subset B_R$. Thus $\Gamma = O((kR)^4)$ uniformly, even for disconnected and irregular objects or size-dependent material patterns. The extra power follows from applying one coercive Maxwell inverse to the two incident fields. For a fixed rescaled object we identify the quartic coefficient through two exact electrostatic response moments and prove an $O((kR)^6)$ remainder. Two actual dielectric spheres with radii $R/20, R/10$, center separation R, and relative permittivities $1+\mathrm{i}, 1+2\mathrm{i}$ have a coefficient C satisfying $36/1299125 \le C \le 44/1295875$ and $\left|\Gamma - C(kR)^4\right| \le 21(kR)^6/2000$ for $kR \le 1/4$. This establishes quartic exponent attainment without identifying the optimal prefactor. A second, homogeneous-material sphere certificate establishes exponent sharpness for every numerical pair $0 < m \le M$. Full-vector spherical-wave calculations corroborate the result. Prior work already establishes extinction symmetry, cross-source constraints, and unequal-sphere scattering asymmetry; scientific priority for the precise uniform theorem is not certified.

Contents

1 Question, assumptions, and principal results **2**

2 One Maxwell problem for both sources **4**

3 Uniform Maxwell estimates on measurable supports **4**
3.1 Proof of the uniform quartic theorem 6

4 The exact quartic coefficient for a fixed rescaled object **6**

5 A finite two-sphere coefficient certificate **7**
5.1 Exact electrostatic interaction control 7
5.2 Exact isolated-sphere moment identities 8
5.3 Noncancellation for the actual interacting spheres 8
5.4 An explicit full-Maxwell remainder 9
5.5 Passive causal material realization 10

6 Homogeneous unequal spheres: every fixed loss and cap **10**
6.1 Exact first interaction and its sign 10
6.2 All-multipole error enclosure and resource-class sharpness 11

7 Full-vector numerical verification **12**
7.1 Method and all-channel accounting 12
7.2 Results and convergence 13

8 Prior work and scientific positioning **14**

9 Adversarial audit, scope, and preserved research record **15**

1 Question, assumptions, and principal results

Miller and Monticone's OQ7 asks for bounds on the difference between absorption and scattering under time-reversed illumination when reciprocity fixes their sum [1]. This manuscript addresses one precise shrinking-object version. All cross sections are those of unit-amplitude free-space plane waves. No finite-power port normalization, collection cone, substrate, periodic continuation, or dipole model is used in the theorems.

Use time dependence $e^{-\mathrm{i}\omega t}$, vacuum wavenumber $k > 0$, relative permeability I, and relative permittivity $I + \chi$ on a measurable set $D \subset B_R(0)$ of positive volume V. The susceptibility is zero outside D. Almost everywhere in D, require

$$\chi^T = \chi, \qquad \|\chi\|_{\mathrm{op}} \le M, \qquad B := \operatorname{Im}_H \chi = \frac{\chi - \chi^\dagger}{2\mathrm{i}} \succeq mI, \qquad 0 < m \le M < \infty. \tag{1}$$

The constants m, M are fixed across the class. No sign condition on $\operatorname{Re}_H \chi$ is necessary. In particular, bounded negative real contrast is allowed. The material law is local and contains only electric dielectric response. Emergent magnetic multipoles of dielectric currents remain part of the exact Maxwell problem.

Let $d, e \in \mathbb{R}^3$ be unit vectors with $d \cdot e = 0$. The incident electric fields are

$$f_\pm(x) = e\, e^{\pm \mathrm{i}kd\cdot x}. \tag{2}$$

The magnetic fields are $Z_0 H_{\mathrm{inc},\pm} = \pm(d \times e)e^{\pm \mathrm{i}kd\cdot x}$. This is the correct time-reversed pair. A constant phase on a linear polarization is immaterial and can be chosen to make e real. For a general elliptical polarization the two constant electric vectors are different, and the theorem below does not apply.

Write $A_\pm, S_\pm, X_\pm$ for absorption, total scattering, and extinction cross sections. Reciprocity gives $X_+ = X_- = X > 0$, and define

$$\Gamma = \frac{|A_+ - A_-|}{X} = \frac{|S_+ - S_-|}{X}. \tag{3}$$

Theorem 1.1 (Uniform quartic upper bound)**.** *Every susceptibility and support satisfying (1) has a unique outgoing locally finite-energy weak Maxwell solution. Put*

$$a_v = \left(\frac{3V}{4\pi}\right)^{1/3}, \qquad \rho = \frac{M^2}{m}, \qquad L_k = m^{-1} + 1 + \frac{3}{4}k^2 a_v^2. \tag{4}$$

For the same scatterer driven by (2),

$$\boxed{\Gamma \le \min\left\{1, \frac{2}{3\pi}\rho^2 L_k k^4 R V\right\}.} \tag{5}$$

In particular, with $t = kR$,

$$\boxed{\Gamma \le \min\left\{1, \frac{8}{9}\rho^2\left(m^{-1} + 1 + \frac{3}{4}t^2\right)t^4\right\}.} \tag{6}$$

These are exact inequalities at every $k, R > 0$. The $O(t^4)$ constant is uniform over all measurable supports, all numbers of components, and all admissible material patterns, including patterns that vary with R.

Theorem 1.2 (Finite-sphere attainment with a quantitative remainder)**.** *At fixed k, take spheres centered at $\mp Re_z/2$, with radii $R/20$ and $R/10$, and set $\chi_1 = \mathrm{i}I$, $\chi_2 = 2\mathrm{i}I$, respectively. Illuminate with $e = e_x, d = e_z$. They lie inside B_R and satisfy (1) with $m = 1, M = 2$. The exact Maxwell cross sections obey*

$$\Gamma(t) = Ct^4 + O(t^6), \qquad \frac{36}{1299125} \le C \le \frac{44}{1295875}, \tag{7}$$

and, more explicitly,

$$\boxed{\left|\Gamma(t) - Ct^4\right| \le \frac{21}{2000}t^6 \quad (0 < t \le 1/4), \qquad \Gamma(t) \ge \frac{27}{1299125}t^4 \quad (0 < t \le 1/40).} \tag{8}$$

The two physical experiments use the same spatial susceptibility.

Theorem 1.3 (Sharp exponent for every fixed resource class)**.** *For every $0 < m \le M$, the supremum of Γ over (1) with $D \subset B_R$ satisfies*

$$\sup_{D,\chi} \Gamma = \Theta_{m,M}((kR)^4) \qquad \text{as } kR \downarrow 0. \tag{9}$$

The matching lower family can use the single homogeneous scalar susceptibility $\chi = \mathrm{i}mI$ on two unequal disjoint spheres. Here $\Theta_{m,M}$ means that positive constants c, C, t_0, depending only on the fixed resource parameters, satisfy $ct^4 \le \sup\Gamma \le Ct^4$ for $0 < t < t_0$. Any fixed real transverse incident pair is covered by rotating the lower construction.

Section 6 proves Theorem 1.3. Theorems 1.1–1.3 establish sharpness of the quartic *exponent*. They do not establish an optimal coefficient in (6), an exact finite-t optimum, or sharpness under an additional fixed positive extinction threshold. In this family extinction vanishes with volume. The larger OQ7 problem remains distinct.

Length convention. R in the upper theorem is any specified enclosing-ball radius. In the explicit family it is chosen as the center separation, which is a valid enclosing-radius resource. The object also lies in $B_{3R/5}(0)$. Its *minimum* enclosing radius is $23R/40$, centered at $Re_z/40$. If normalization uses that minimum radius, multiply C by $(40/23)^4$. This convention differs from the diameter convention used in parts of the earlier fixed-resource manuscript.

2 One Maxwell problem for both sources

On $\mathcal{H} = L^2(D;\mathbb{C}^3)$ use $\langle u, v\rangle = \int_D \overline{u}\cdot v$ and the bilinear pairing $[u, v] = \int_D u \cdot v$. With $p = \chi E$ and zero extension of p, define

$$\mathcal{K}_k p = (k^2 I + \nabla\nabla\cdot)(g_k * p)|_D, \qquad g_k(x) = \frac{e^{\mathrm{i}k|x|}}{4\pi\,|x|}. \tag{10}$$

Derivatives are distributional. In particular, the electrostatic contact term is retained. The volume equation and response operator are

$$\mathcal{L}_k p = f, \qquad \mathcal{L}_k = \chi^{-1} - \mathcal{K}_k, \qquad \mathcal{T}_k = \mathcal{L}_k^{-1}, \qquad p_\pm = \mathcal{T}_k f_\pm. \tag{11}$$

The exact joint design problem at radius R is therefore the supremum of (3) over D, χ, p_+, p_- subject to (1), $D \subset B_R$, and both equations in (11) with the *same* local multiplier χ. This keeps full reactive and radiative Maxwell compatibility. It is not replaced by independent power constraints.

For clarity, let $F : \mathbb{C}^2 \to \mathcal{H}$ and $P : \mathbb{C}^2 \to \mathcal{H}$ have columns $f_\pm$ and $p_\pm$. Define the positive multiplication operator

$$W = -\operatorname{Im}_H \chi^{-1} = \chi^{-\dagger} B \chi^{-1}. \tag{12}$$

The two-source power identity is the matrix identity

$$\operatorname{Im}_H(F^\dagger P) = P^\dagger (W + \operatorname{Im}_H \mathcal{K}_k) P. \tag{13}$$

Reciprocity additionally gives $[f_i, p_j] = [f_j, p_i]$. These off-diagonal identities are automatic consequences of the joint Maxwell equation. Neither they nor the diagonal power identities are asserted to be a sufficient local constitutive realizability criterion by themselves. Cross-source and shared-structure constraints have established precedents in electromagnetic optimization [6, 7]; the argument here retains the complete common inverse rather than optimizing a relaxed pair of currents.

3 Uniform Maxwell estimates on measurable supports

Lemma 3.1 (Bounded dyadic Green operator)**.** *On $\mathcal{H}$, with a_v as in (4),*

$$\|\mathcal{K}_0\| \le 1, \qquad \|\mathcal{K}_k - \mathcal{K}_0\| \le \frac{3}{4} k^2 a_v^2. \tag{14}$$

Proof. On $L^2(\mathbb{R}^3;\mathbb{C}^3)$ the static Fourier multiplier is $-\xi\xi^T/|\xi|^2$, of norm one. Extension and restriction are contractions, proving the first claim on arbitrary measurable D.

The difference has an ordinary locally integrable kernel. Set $r = |x - y|$, $u = kr$, and $b(u) = e^{\mathrm{i}u}(\mathrm{i}u - 1) + 1$. In the orthonormal radial/tangential frame its eigenvalues are

$$\frac{-2b(u)}{4\pi r^3}, \qquad \frac{u^2 e^{\mathrm{i}u} + b(u)}{4\pi r^3}. \tag{15}$$

Since $b(0) = 0$ and $b'(u) = -ue^{\mathrm{i}u}$, $|b(u)| \le u^2/2$. Thus the kernel norm is at most $3k^2/(8\pi r)$. Its distributional contact terms have canceled. Rearrangement of the radially decreasing function $1/r$ gives

$$\sup_x \int_D \frac{\mathrm{d}y}{|x - y|} \le 2\pi a_v^2,$$

and the same bound with x, y interchanged. Schur's test proves the second claim. No boundary trace, component count, or minimum separation occurs. □

Lemma 3.2 (Complete radiation and the optical theorem)**.** *Define*

$$(\mathcal{F}_k p)(n) = \Pi_n \int_D p(x) e^{-\mathrm{i}kn\cdot x}\,\mathrm{d}x, \qquad \Pi_n = I - nn^T, \quad n \in S^2. \tag{16}$$

Then

$$\langle p, (\mathrm{Im}_H\, \mathcal{K}_k) p\rangle = \frac{k^3}{16\pi^2}\,\|\mathcal{F}_k p\|^2_{L^2(S^2)} \geq 0, \tag{17}$$

$$\|\mathcal{F}_k\|^2 \leq \frac{8\pi}{3} V. \tag{18}$$

For unit incident electric amplitude, the exact cross sections are

$$X = k\,\mathrm{Im}\,\langle f, p\rangle\,, \qquad A = k\,\langle p, Wp\rangle = k\int_D E^\dagger B E, \qquad S = \frac{k^4}{16\pi^2}\,\|\mathcal{F}_k p\|^2\,, \qquad X = A + S. \tag{19}$$

Proof. Use the identity $\sin(kr)/(4\pi r) = k(16\pi^2)^{-1}\int_{S^2} e^{\mathrm{i}kn\cdot x}\,\mathrm{d}\Omega$. Applying $k^2 I + \nabla\nabla\cdot$ inserts $k^2\Pi_n$, proving (17). The identity first holds for smooth compact currents; approximation of their zero extensions in an ambient ball proves it for all $p \in L^2(D)$, including sets with empty interior.

Pointwise Cauchy–Schwarz gives $|\mathcal{F}_k p(n)|^2 \leq V\int_D |\Pi_n p(x)|^2\,\mathrm{d}x$. Integrate and use $\int_{S^2}\Pi_n\,\mathrm{d}\Omega = (8\pi/3)I$ to obtain (18). The outgoing far-field amplitude is $k^2\mathcal{F}_k p/(4\pi)$; hence its complete squared norm is the stated scattering cross section. Taking the imaginary part of the volume equation gives $X = A + S$ and the usual forward optical theorem. □

Lemma 3.3 (Coercive common inverse and weak Maxwell equivalence)**.** *Under* (1)*,* $W \succeq \rho^{-1} I$*,* $\|\mathcal{T}_k\| \leq \rho$*, and* $\|\mathcal{L}_k\| \leq L_k$*. The volume solution is the unique outgoing weak Maxwell solution.*

Proof. For every vector v, $|\chi v| \geq m\,|v|$ by the loss floor, so $\|\chi^{-1}\| \leq m^{-1}$. Equation (12) gives $v^\dagger W v \geq m\,|\chi^{-1}v|^2 \geq (m/M^2)\,|v|^2$. Using (17),

$$-\,\mathrm{Im}\,\langle p, \mathcal{L}_k p\rangle = \langle p, Wp\rangle + \langle p, (\mathrm{Im}_H\,\mathcal{K}_k)p\rangle \geq \rho^{-1}\,\|p\|_2^2\,. \tag{20}$$

The bounded sesquilinear form of $\mathrm{i}\mathcal{L}_k$ is coercive. Complex Lax–Milgram supplies existence, surjectivity, uniqueness, and the inverse-norm bound. Lemma 3.1 gives the asserted bound on $\mathcal{L}_k$.

For completeness, $g_k * p \in H^2_{\mathrm{loc}}$ by local Helmholtz regularity for a compactly supported L^2 source. Consequently the constructed $E = f + (k^2 I + \nabla\nabla\cdot)(g_k * p)$ is locally L^2, and $\nabla\times E = \nabla\times f + k^2\nabla\times(g_k * p)$ is locally H^1. Thus $E \in H_{\mathrm{loc}}(\mathrm{curl})$, the magnetic field obtained from Faraday's law also has the required local regularity, and the full Maxwell equations hold distributionally with the specified material coefficient. The volume potential is outgoing. Conversely, subtracting both the incident field and the unrestricted volume potential from any locally finite-energy outgoing weak solution leaves an entire outgoing vacuum Maxwell field. Vacuum uniqueness makes it zero. No geometrical boundary regularity was used. □

Lemma 3.4 (Exact reciprocal extinction)**.** *For* (2)*,* $X_+ = X_- > 0$.

Proof. The dyadic kernel is symmetric under transpose and simultaneous exchange of its two arguments. Hence $\mathcal{L}_k$ and its inverse are symmetric for the bilinear pairing. Since $f_- = \overline{f_+}$,

$$\langle f_+, \mathcal{T}_k f_+\rangle = [f_-, \mathcal{T}_k f_+] = [f_+, \mathcal{T}_k f_-] = \langle f_-, \mathcal{T}_k f_-\rangle\,. \tag{21}$$

Their imaginary parts give equality. Since $V > 0$, $f_\pm \neq 0$ in $\mathcal{H}$; invertibility implies $p_\pm \neq 0$ and $X \geq k\rho^{-1}\,\|p_\pm\|^2 > 0$. □

3.1 Proof of the uniform quartic theorem

The crucial same-material estimate is

$$\|p_+ - p_-\|_2 \le \rho \, \|f_+ - f_-\|_2 \le 2\rho k R\sqrt{V}. \tag{22}$$

Put $h = (\rho X/k)^{1/2}$. Passivity and common extinction give $\|p_\pm\|_2 \le h$, while the equation itself gives

$$\sqrt{V} = \|f_+\|_2 = \|\mathcal{L}_k p_+\|_2 \le L_k \, \|p_+\|_2 \le L_k h. \tag{23}$$

The exact radiation norm and the difference of two quadratic forms yield

$$\begin{aligned} |S_+ - S_-| &\le \frac{k^4 V}{6\pi} \, \|p_+ - p_-\|_2 \, (\|p_+\|_2 + \|p_-\|_2) \\ &\le \frac{2}{3\pi} \rho^2 L_k k^4 R V \, X. \end{aligned} \tag{24}$$

Divide by X and use $0 \le A_\pm \le X$ to prove (5). The enclosing-ball facts $V \le 4\pi R^3/3$ and $a_v \le R$ give (6). This proves Theorem 1.1.

The proof needs no imposed extinction floor or lower volume fraction. The factor $\sqrt{V}$ in (23) is the norm of the very same incident field on the very same support. It remains valid when V/R^3 tends to zero. The old independent-current estimate controls each scattering fraction at cubic order; (22) is the extra compatibility estimate that supplies one additional power.

4 The exact quartic coefficient for a fixed rescaled object

Now fix a positive-volume measurable $\Omega \subset B_1$ and a material pattern $\chi(y)$ satisfying (1). Let the physical body be $D_R = R\Omega$, with $\chi_R(x) = \chi(x/R)$, and put $t = kR$. All operators in this section act on $L^2(\Omega; \mathbb{C}^3)$ with dimensionless wavenumber t. Define

$$q_0 = \mathcal{T}_0 e, \qquad q_1 = \mathcal{T}_0[(d \cdot y)e], \qquad p_0 = \int_\Omega q_0, \quad p_1 = \int_\Omega q_1, \qquad D_0 = \operatorname{Im}(e \cdot p_0). \tag{25}$$

Here q_j are polarization densities and p_j are their total moments. The affine input defining q_1 is the first term of a Maxwell plane-wave expansion; it is not required to be a curl-free electrostatic applied field. The bounded volume resolvent acts on all L^2 inputs.

Theorem 4.1 (Full-vector coefficient formula)**.** *The number D_0 is strictly positive. For the fixed rescaled pattern,*

$$\boxed{\Gamma(t) = \frac{2\left|\operatorname{Im}(p_0^\dagger p_1)\right|}{3\pi D_0} t^4 + O(t^6).} \tag{26}$$

If the numerator vanishes, the conclusion is $\Gamma = O(t^6)$ for that pattern; a nonzero coefficient is not presumed.

Proof. The resolvent identity and Lemmas 3.1–3.3 give

$$\|\mathcal{T}_t - \mathcal{T}_0\| \le \frac{3}{4} \rho^2 a_{v,\Omega}^2 t^2. \tag{27}$$

Set $c_t = e\cos(td \cdot y)$, $s_t = e\sin(td \cdot y)$ and $u_t = \mathcal{T}_t c_t$, $v_t = \mathcal{T}_t s_t$. Taylor inequalities on the bounded support and (27) imply

$$q_\pm(t) = u_t \pm \mathrm{i} v_t, \qquad u_t = q_0 + O_{L^2}(t^2), \qquad v_t = t q_1 + O_{L^2}(t^3). \tag{28}$$

These are expansions of the exact Maxwell solutions.

Normalize the all-angle radiation operator by

$$Q_t = \frac{1}{16\pi^2}\mathcal{F}_t^\dagger \mathcal{F}_t, \qquad Q_t(x,y) = \frac{1}{16\pi^2}\int_{S^2} \Pi_n e^{\mathrm{i}tn\cdot(x-y)}\,\mathrm{d}\Omega_n. \tag{29}$$

Changing n to $-n$ proves that its kernel is real and even in t. For diameter L_Ω and volume W_Ω, the elementary cosine remainder gives the useful explicit estimate

$$\|Q_t - Q_0\| \le \frac{W_\Omega L_\Omega^2}{8\pi} t^2, \qquad \langle u, Q_0 v\rangle = \frac{1}{6\pi}\left(\int_\Omega u\right)^\dagger \int_\Omega v. \tag{30}$$

It follows that

$$\begin{aligned}\frac{S_+ - S_-}{k^4 R^6} &= -4\,\mathrm{Im}\,\langle u_t, Q_t v_t\rangle \\ &= -\frac{2}{3\pi}\,t\,\mathrm{Im}(p_0^\dagger p_1) + O(t^3).\end{aligned} \tag{31}$$

The observation-direction phase has not been discarded. Its odd term cancels in the full angular kernel, before any estimate is made.

By bilinear symmetry of $\mathcal{T}_t$ the mixed sine–cosine terms in the forward work cancel exactly:

$$\frac{X}{kR^3} = \mathrm{Im}\{[c_t, \mathcal{T}_t c_t] + [s_t, \mathcal{T}_t s_t]\} = D_0 + O(t^2). \tag{32}$$

At $t = 0$, $\mathcal{K}_0$ is self-adjoint, so $D_0 = \langle q_0, W q_0\rangle > 0$. Dividing the absolute value of (31) by (32), using $k^3R^3 = t^3$, proves (26). □

The expansion treats every radiating component of the current. Electric quadrupoles, magnetic dipoles, higher multipoles, and all their interference terms are contained in the exact Q_t and in its norm remainder. There is no assumption that the physical current is uniform inside a particle. Translating the reference origin changes p_1 by a real multiple of p_0 and therefore leaves $\mathrm{Im}(p_0^\dagger p_1)$ unchanged, as required physically.

5 A finite two-sphere coefficient certificate

The geometry in Theorem 1.2 is now fixed in reference coordinates:

$$\Omega_1 = B_a(-e_z/2), \qquad \Omega_2 = B_{2a}(e_z/2), \qquad a = \frac{1}{20}, \quad \chi_1 = \mathrm{i}I, \quad \chi_2 = 2\mathrm{i}I. \tag{33}$$

Let $W_\Omega = |\Omega| = 12\pi a^3$; below write this scalar as w to distinguish it from the material loss operator W. The entire geometry lies in $B_{3/5}$ and its intersphere surface gap is $1 - 3a = 17/20$.

5.1 Exact electrostatic interaction control

Split $\mathcal{K}_0$ into within-sphere and between-sphere blocks. Put

$$A_b = \mathrm{diag}\{(\chi_j^{-1} - \mathcal{K}_{jj})^{-1}\}_{j=1,2}, \qquad C_{\mathrm{off}} = (\mathcal{K}_{ij})_{i\neq j}. \tag{34}$$

For these two materials $-\mathrm{Im}_H\,\chi_j^{-1} \succeq I/2$, so $\|A_b\| \le 2$. Between distinct spheres the exact static kernel is

$$H(r) = \frac{3rr^T - |r|^2 I}{4\pi\,|r|^5}, \qquad \|H(r)\|_{\mathrm{op}} = \frac{1}{2\pi\,|r|^3}. \tag{35}$$

Cauchy–Schwarz in each block gives

$$\|C_{\mathrm{off}}\| \le c := \frac{\sqrt{|\Omega_1|\,|\Omega_2|}}{2\pi(1-3a)^3} = \frac{4\sqrt{2}}{14739} < \frac{1}{2500}. \tag{36}$$

The block-off-diagonal operator has norm equal to the larger block norm; there is no extra factor of two. The Neumann resolvent identity thus gives

$$\delta := \|\mathcal{T}_0 - A_b\| \le \frac{4c}{1-2c} < \frac{1}{600}. \tag{37}$$

This bounds the entire interacting electrostatic operator. It sums all multiple reflections and includes all static multipoles. The comparison with isolated spheres is not a point-dipole approximation imposed on the physical solution.

5.2 Exact isolated-sphere moment identities

For a scalar susceptibility in a ball, the response to a constant vector is the constant polarization $\beta_j e$, where $\beta_j = 3\chi_j/(3+\chi_j)$. Bilinear symmetry gives, for every vector test v and arbitrary $h \in L^2(\Omega_j;\mathbb{C}^3)$,

$$v \cdot \int_{\Omega_j} \mathcal{T}_j h = [v, \mathcal{T}_j h] = [\mathcal{T}_j v, h] = \beta_j v \cdot \int_{\Omega_j} h.$$

Therefore

$$\int_{\Omega_j} \mathcal{T}_j h = \beta_j \int_{\Omega_j} h. \tag{38}$$

For the affine drive $h = z e_x$, this is exactly the center coordinate times the constant-drive sphere dipole. Equivalently, the centered odd part of h has zero total moment by inversion parity. This justifies the moment even though the affine input has a curl.

The two isolated polarizabilities are

$$\alpha_1 = 4\pi a^3 \frac{1+3\mathrm{i}}{10}, \qquad \alpha_2 = 32\pi a^3 \frac{4+6\mathrm{i}}{13}. \tag{39}$$

For the block comparison,

$$p_0^b = (\alpha_1+\alpha_2)e_x, \qquad p_1^b = \tfrac{1}{2}(\alpha_2-\alpha_1)e_x. \tag{40}$$

Exact rational arithmetic yields

$$I_b := \mathrm{Im}((p_0^b)^\dagger p_1^b) = -\frac{8}{195}w^2, \qquad D_b := \mathrm{Im}(e_x \cdot p_0^b) = \frac{173}{130}w. \tag{41}$$

5.3 Noncancellation for the actual interacting spheres

Let $b = 3/5$, so $|z| \le b$ on Ω. The integration-map norm is $\sqrt{w}$, hence (37) gives

$$\begin{aligned} \left|p_0 - p_0^b\right| &\le \delta w, & \left|p_1 - p_1^b\right| &\le b\delta w, \\ \left|p_0^b\right| &\le 2w, & \left|p_1^b\right| &\le 2bw. \end{aligned} \tag{42}$$

Writing $I_0 = \mathrm{Im}(p_0^\dagger p_1)$, these imply

$$|I_0 - I_b| \le b(4\delta+\delta^2)w^2 < \frac{3}{5}\left(\frac{4}{600} + \frac{1}{600^2}\right)w^2 < \frac{1}{10}\,|I_b|\,. \tag{43}$$

The last relative upper estimate is the rational number 0.097540625, strictly below 0.1. In particular $I_0 < 0$ and

$$\frac{12}{325}w^2 \le |I_0| \le \frac{44}{975}w^2, \qquad \frac{10367}{7800}w \le D_0 \le \frac{10393}{7800}w. \tag{44}$$

Substitute these inequalities in Theorem 4.1:

$$\frac{2304}{10393}a^3 \le C \le \frac{2816}{10367}a^3. \tag{45}$$

At $a = 1/20$ this is (7), namely

$$2.77109592995 \times 10^{-5} \le C \le 3.39538921578 \times 10^{-5}. \tag{46}$$

The isolated comparison gives $C_b = (128/519)a^3 = 3.08285163776 \times 10^{-5}$. It is not the exact interacting coefficient.

5.4 An explicit full-Maxwell remainder

Use the conservative reference diameter $L = 6/5$ and $0 < t \le 1$. The tangential and radial eigenvalues of the dynamic-kernel difference are $(4\pi r^3)^{-1}$ times

$$A(s) = e^{\mathrm{i}s}(s^2 + \mathrm{i}s - 1) + 1, \qquad B(s) = 2\{e^{\mathrm{i}s}(1 - \mathrm{i}s) - 1\}, \qquad s = tr.$$

Their derivatives are $A'(s) = s(1+\mathrm{i}s)e^{\mathrm{i}s}$ and $B'(s) = 2se^{\mathrm{i}s}$. Since $s \le 6/5 < \sqrt{3}$, $|A(s)|, |B(s)| \le s^2$ by integration. Thus the kernel norm is at most $t^2/(4\pi r)$. Integrating over an enclosing radius-L ball about each source point and applying Schur's test gives

$$\|\mathcal{K}_t - \mathcal{K}_0\| \le \frac{18}{25}t^2, \qquad \|\mathcal{T}_t\| \le 2, \qquad \|\mathcal{T}_t - \mathcal{T}_0\| \le \frac{72}{25}t^2. \tag{47}$$

Here the exact positive radiation term preserves loss coercivity for every real $t \ge 0$; no artificial radiation damping is introduced.

Equations (28)–(30) now have explicit constants:

$$\begin{aligned} \|u_t - q_0\| &\le \frac{81}{25}t^2\sqrt{w}, & \|v_t - tq_1\| &\le \frac{9}{5}t^3\sqrt{w}, \\ \|u_t\| &\le 2\sqrt{w}, & \|v_t\| &\le 2bt\sqrt{w}, \\ \|Q_0\| &= \frac{w}{6\pi}, & \|Q_t - Q_0\| &\le \frac{wL^2}{8\pi}t^2. \end{aligned} \tag{48}$$

The first two follow from $|\cos(tz) - 1| \le t^2b^2/2$ and $|\sin(tz) - tz| \le t^3b^3/6$.

Set $\mathscr{S}(t) = (S_+ - S_-)/(k^4R^6)$, $\mathscr{S}_0 = -2I_0/(3\pi)$, and $\mathscr{D}(t) = X/(kR^3)$. Splitting the bilinear radiation error into the changes in u, v, Q gives

$$|\mathscr{S}(t) - t\mathscr{S}_0| \le \frac{168}{25\pi}w^2t^3, \tag{49}$$

$$|\mathscr{D}(t) - D_0| \le \frac{108}{25}wt^2. \tag{50}$$

For example, to verify (49), write

$$\langle u_t, Q_tv_t\rangle - t\langle q_0, Q_0q_1\rangle = \langle u_t, (Q_t - Q_0)v_t\rangle + \langle u_t - q_0, Q_0v_t\rangle + \langle q_0, Q_0(v_t - tq_1)\rangle.$$

The three constants before multiplying by four are $54/(125\pi)$, $81/(125\pi)$, and $3/(5\pi)$, times w^2t^3. Their sum is $42/(25\pi)$. Equation (50) follows by applying the same norm bounds to the two terms of (32).

For $t \le 1/4$, (44) and (50) give $\mathscr{D}(t) \ge w$ and $D_0 \ge w$. Also $|\mathscr{S}_0| \le 88w^2/(2925\pi)$. Therefore

$$\left|\Gamma/t^4 - C\right| \le \left\{\frac{168}{25} + \frac{88}{2925}\frac{108}{25}\right\}\frac{w}{\pi}t^2 < \frac{7w}{\pi}t^2. \tag{51}$$

Since $w = 12\pi/8000$, this is the remainder in (8). At $t \le 1/40$, the maximum remainder relative to t^4 is less than one quarter of the lower endpoint in (7). This proves the stated positive finite-t lower bound and completes Theorem 1.2.

5.5 Passive causal material realization

This construction is a size limit at a fixed operating frequency $\omega_0 > 0$, rather than a claim of frequency-independent loss as $\omega \to 0$. For any fixed $\gamma > 0$, the passive Lorentz laws

$$\chi_j(\omega) = \frac{j\gamma\omega_0}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega}\, I, \qquad j = 1, 2, \tag{52}$$

have poles in the lower half plane and positive imaginary part for $\omega > 0$. At ω_0 they give exactly $\chi_j = \mathrm{i}jI$. The geometry shrinks while these material functions remain fixed.

6 Homogeneous unequal spheres: every fixed loss and cap

The explicit heterogeneous example has a comparatively large coefficient and a convenient rational certificate. The exponent is also attained when both spheres have the same scalar susceptibility. This additionally settles the potentially degenerate resource class $M = m$.

Fix $\mu > 0$ and use $\chi = \mathrm{i}\mu I$ on both balls, centered at $\mp e_z/2$, of radii $a, 2a$, where $0 < a \le 1/20$ will be fixed below. Put $v = 4\pi a^3/3$, $v_1 = v$, $v_2 = 8v$, $w = 9v$, $h = 1 - 3a \ge 4/5$, $b = 3/5$, and

$$\beta = \frac{3\mathrm{i}\mu}{3 + \mathrm{i}\mu}, \qquad |\beta| \le \min\{\mu, 3\}, \qquad \operatorname{Im}\beta = \frac{9\mu}{9 + \mu^2} > 0. \tag{53}$$

Write $A = \operatorname{diag}(A_1, A_2)$ for the isolated electrostatic resolvents, C_o for the off-diagonal static Green blocks, and T for the exact two-sphere electrostatic response. Coercivity gives $\|A\|, \|T\| \le \mu$. The kernel estimate gives

$$\|C_o\| \le \frac{4\sqrt{2}a^3}{3h^3} \le 4a^3. \tag{54}$$

Iterating the exact resolvent identity twice yields

$$T = A + AC_oA + AC_oAC_oT, \qquad \|T - A - AC_oA\| \le \mu^3 \|C_o\|^2. \tag{55}$$

This identity uses the actual inverse throughout and needs no Neumann series convergence assumption.

6.1 Exact first interaction and its sign

For $f = e_x$ and $g = ze_x$, write the corresponding moments through the first interaction as $\widetilde{p}_j = p_j^b + p_j^{(1)}$. The block moments are

$$p_0^b = \beta w e_x, \qquad p_1^b = \tfrac{1}{2}\beta(v_2 - v_1)e_x.$$

Every entry of $H(x-y)$ is harmonic away from $x=y$. Applying the ball mean-value property twice gives the exact identity

$$\int_{\Omega_i}\int_{\Omega_j} H(x-y)\,\mathrm{d}y\,\mathrm{d}x = v_i v_j H(c_i-c_j), \qquad i\neq j. \tag{56}$$

Since $H(e_z)e_x = g_0 e_x$, $g_0=-1/(4\pi)$, the first correction to the constant-drive total moment is exactly

$$p_0^{(1)} = 2g_0\beta^2 v_1 v_2 e_x, \qquad |\widetilde{p}_0| \le |\beta|\, w\left(1+\frac{16}{27}\,|\beta|\,a^3\right) \le 2\,|\beta|\,w. \tag{57}$$

For the affine drive on sphere j, decompose $A_j g = z_j\beta e_x + q_j^o$, with $q_j^o = A_j[(z-z_j)e_x]$. Equation (38) gives $\int q_j^o = 0$, and spherical second moments give

$$\left\|q_j^o\right\| \le \mu r_j\sqrt{v_j/5}, \qquad \int_{\Omega_j} |y-c_j|\left|q_j^o(y)\right|\,\mathrm{d}y \le \frac{\mu\sqrt{3}}{5} r_j^2 v_j. \tag{58}$$

The uniform center terms in $p_1^{(1)}$ cancel because $z_1+z_2=0$. First integrate the receiving-ball response by (38), then apply its harmonic mean-value property:

$$p_1^{(1)} = \beta\sum_{i\neq j} v_i \int_{\Omega_j} H(c_i-y)q_j^o(y)\,\mathrm{d}y.$$

Use $\int q_j^o = 0$ to subtract $H(c_i-c_j)$. Direct differentiation of the tensor kernel, for a real unit vector u and $n = r/|r|$, gives

$$DH(r)[u] = \frac{3(un^T+nu^T)+3(n\cdot u)I-15(n\cdot u)nn^T}{4\pi\,|r|^4}, \qquad \|DH(r)[u]\| \le \frac{6}{\pi\,|r|^4}. \tag{59}$$

The intervening segments stay at distance at least h from zero. Consequently the *entire* affine correction, including every electrostatic multipole, satisfies

$$\left|p_1^{(1)}\right| \le \frac{6\sqrt{3}\,|\beta|\,\mu}{\pi h^4} a^2 v_1 v_2. \tag{60}$$

The imaginary part of $(p_0^b)^\dagger p_1^b$ is zero, whereas the exact first constant-drive interaction produces the positive number

$$\mathscr{L} := \mathrm{Im}((p_0^{(1)})^\dagger p_1^b) = -g_0 v_1 v_2 (v_2-v_1)\,|\beta|^2\,\mathrm{Im}\,\beta = \frac{2688\pi^2\mu^3}{(9+\mu^2)^2}a^9 > 0. \tag{61}$$

Thus a homogeneous material does not impose a cancellation of the quartic coefficient for unequal interacting spheres.

6.2 All-multipole error enclosure and resource-class sharpness

Equations (57)–(60) imply

$$\frac{\left|\mathrm{Im}(\widetilde{p}_0^\dagger\widetilde{p}_1)-\mathscr{L}\right|}{\mathscr{L}} \le \frac{48\sqrt{3}}{7h^4}(9+\mu^2)a^2 < 30(9+\mu^2)a^2. \tag{62}$$

For the remaining interactions let $\delta_2 = \mu^3 \|C_o\|^2$. Equation (55) gives $|p_0 - \widetilde{p}_0| \le \delta_2 w$ and $|p_1 - \widetilde{p}_1| \le b\delta_2 w$. Use one exact factor, $|p_1| \le \mu b w$, in the difference of cross products, rather than omitting a second-order error:

$$\begin{aligned}\left|\mathrm{Im}(p_0^\dagger p_1) - \mathrm{Im}(\widetilde{p}_0^\dagger \widetilde{p}_1)\right| &\le b\delta_2(\mu + 2|\beta|)w^2 \\ &\le \frac{9}{5}\mu^4 \|C_o\|^2 w^2. \end{aligned} \tag{63}$$

Combining the preceding bounds gives the completely explicit certificate

$$\boxed{\frac{\left|\mathrm{Im}(p_0^\dagger p_1) - \mathscr{L}\right|}{\mathscr{L}} \le 30(9+\mu^2)a^2 + \frac{54}{35}\mu(9+\mu^2)^2 a^3.} \tag{64}$$

Choose once and for all a positive a satisfying

$$a \le \min\left\{\frac{1}{20}, [120(9+\mu^2)]^{-1/2}, \left[\frac{35}{216\mu(9+\mu^2)^2}\right]^{1/3}\right\}. \tag{65}$$

The right side of (64) is then at most $1/2$, so $\mathscr{L}/2 \le \mathrm{Im}(p_0^\dagger p_1) \le 3\mathscr{L}/2$ for the exact finite spheres.

The static energy identity and the inverse bounds give

$$\frac{\mu w}{(1+\mu)^2} \le D_0 = \frac{1}{\mu}\|Tf\|^2 \le \mu w.$$

Theorem 4.1 now yields a physical full-Maxwell family

$$\Gamma = C_h t^4 + O(t^6), \qquad \frac{224\mu^2}{3(9+\mu^2)^2}a^6 \le C_h \le \frac{224\mu^2(1+\mu)^2}{(9+\mu^2)^2}a^6. \tag{66}$$

For example $\mu = 1, a = 1/100$ makes the relative error in (64) at most $5277/175000 < 0.031$. All choices are fixed before $t \to 0$; there is no simultaneous uncontrolled separation and size limit. Setting $\mu = m$ satisfies (1) for every $M \ge m$, including $M = m$. The positive lower coefficient, the controlled fixed-pattern Maxwell remainder, and Theorem 1.1 prove Theorem 1.3.

7 Full-vector numerical verification

The analytic proofs above establish the theorem. The numerical work independently checks normalization, the claimed physical example, and the size scaling. The implementation is included in the source archive.

7.1 Method and all-channel accounting

We solve the finite-sphere Maxwell boundary problem with regular vector spherical waves inside each sphere and outgoing vector spherical waves outside. Both electric and magnetic wave families are retained for every $1 \le \ell \le \ell_{\max}$. For a coaxial isotropic pair under axial plane waves polarized along x, axial symmetry confines the solution exactly to azimuthal orders $m = \pm 1$. This is a symmetry reduction, not a dipole truncation.

Tangential E and H are matched on both sphere boundaries by weighted least squares after column scaling. Internal magnetic fields carry the correct refractive-index factor for $\mu_r = 1$. Fields from each sphere are evaluated in the coordinate system centered on that sphere and then summed

in global coordinates. Radiation includes the translation phases of both centers and all interference terms before its squared norm is integrated over the entire sphere. Gauss–Legendre quadrature in $\cos\theta$ and two exact azimuthal samples integrate the $m = \pm 1$ intensity. Independently, volume quadrature evaluates $k \operatorname{Im} \int f^\dagger \chi E$ and $k \int E^\dagger (\Im\chi) E$.

No outgoing angular cone or individual-dipole scattering sum is used. For validation, single-sphere results are compared with independently implemented analytic Mie coefficients. Finite numerical truncations and floating-point calculations are checks, not proofs of the infinite-dimensional statement.

7.2 Results and convergence

In the table, the center separation is one and the numerical k is therefore $t = kR$. This is the dimensionless representation of shrinking the body at one fixed physical frequency, not a material frequency sweep. Values use $\ell_{\max} = 6$.

t	Γ	Γ/t^4	Local slope to next row
0.400	$7.40724841 \times 10^{-7}$	$2.89345641 \times 10^{-5}$	3.93325
0.200	$4.84876834 \times 10^{-8}$	$3.03048021 \times 10^{-5}$	3.98342
0.100	$3.06551229 \times 10^{-9}$	$3.06551229 \times 10^{-5}$	3.99586
0.050	$1.92144924 \times 10^{-10}$	$3.07431878 \times 10^{-5}$	3.99897
0.025	$1.20176694 \times 10^{-11}$	$3.07652336 \times 10^{-5}$	—

Table 1: Full-vector cross sections for the actual two-sphere example. Local slope is $\log(\Gamma_i/\Gamma_{i+1})/\log(t_i/t_{i+1})$. The $t = 0.4$ row is a numerical check outside the interval of the explicit remainder estimate; the analytic theorem does not rely on that row.

An affine fit of Γ/t^4 against t^2 using the last three rows has intercept approximately $3.07725625 \times 10^{-5}$, well within the analytic interval. This extrapolation is a numerical estimate, not a sharpened rigorous coefficient enclosure.

Across the five $\ell_{\max} = 6$ calculations, the largest relative optical-theorem residual is 1.44×10^{-14} and the largest relative extinction-reciprocity residual is 1.02×10^{-14}. At $t = 0.1$, increasing $\ell_{\max}$ from seven to nine and increasing the boundary quadrature changes Γ/t^4 by 8.2×10^{-13} relatively. The denser run has relative boundary residual 4.2×10^{-13}. Increasing the independent radiation and volume quadratures changes the respective integrals by less than 1.2×10^{-15} and 4.5×10^{-16} relatively. These are observed numerical residuals, not interval error certificates.

The single-sphere extinction and scattering checks agree with the independent Mie formulas to about 1.1×10^{-14} relative error. Identical spheres in the inversion-symmetric configuration give zero directional scattering difference to numerical precision. An unequal lossless pair likewise gives equal total scattering. The latter control tests the code; it is outside the positive-loss theorem.

At very small t, subtracting absorption cross sections of order t in reference-area units to recover their difference of order t^5 causes severe cancellation. The principal numerical observable therefore uses the exact identity $|A_+ - A_-| = |S_+ - S_-|$ and subtracts the total scattering values, while independently recording absorption and energy residuals. The archive retains low-order truncation failures as well as converged runs, so an apparently favorable slope is not the only validation criterion.

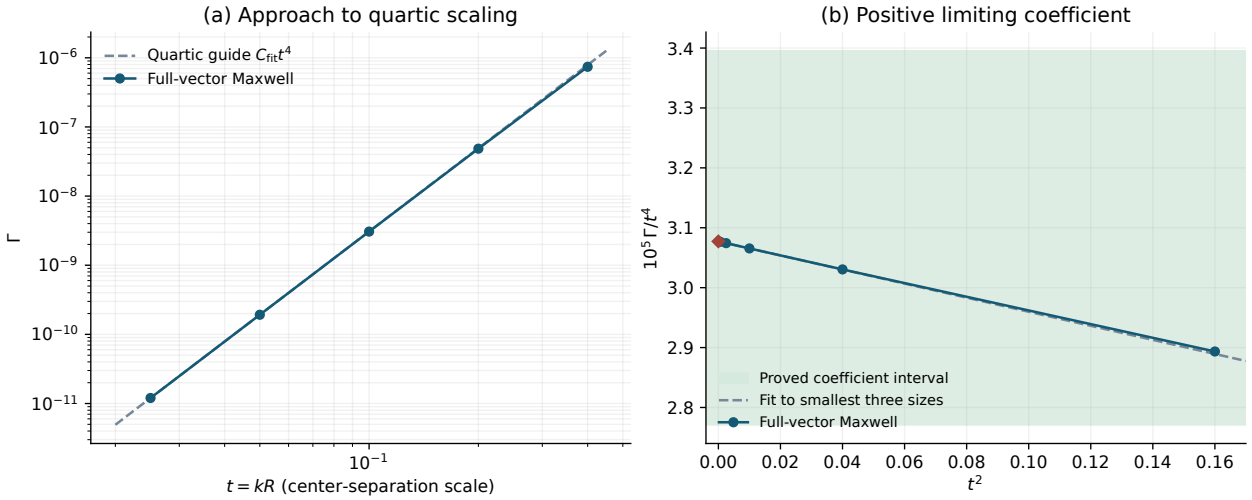


Figure 1: Numerical full-Maxwell scaling of the finite pair. The reference quartic guide uses the numerical extrapolation, and the shaded interval is the independently proved range for the exact coefficient. The proof of sharp exponent does not depend on the fitted guide.

8 Prior work and scientific positioning

Extinction symmetry is an established reciprocity consequence. Sounas and Alù [2] also studied unequal lossy gold nanospheres under opposite illumination, including total scattering and extinction. Thus neither the existence of asymmetric absorption and scattering nor the unequal-sphere example is claimed as a new physical mechanism. The accessible publisher preview and figure captions establish that direct antecedent; a complete page-by-page comparison with their unavailable full derivation remains a priority check before submission.

Heterodimer direction dependence was investigated earlier by Brown et al. [3]. Their measured and simulated collection-cone spectra over a substrate require care when comparing with a whole-space vacuum scattering cross section. Hopkins and coauthors studied absorption/scattering dichroism and its relation to nonorthogonal modes [4, 5]. Their complex-polarization cases also show why the illumination convention cannot be omitted here.

The need for one shared physical structure across multiple incident fields is established in T-operator and cross-correlation approaches [6, 7]. The present theorem uses that requirement in the particularly simple form $p_+ - p_- = \mathcal{T}_k(f_+ - f_-)$. Material-response bounds and radiation-to-loss tradeoffs also predate this work [8, 9]. The preceding OQ7 fixed-resource manuscript already used such estimates to obtain a cubic small-volume bound when the two currents were relaxed separately.

Small-inclusion full-Maxwell expansions have a substantial literature, including Ammari and Khelifi [10]. Coupled-dipole optical theorems and radiation corrections are likewise well established [11, 12]. A point-particle model alone is not a remainder estimate for a finite cluster whose entire separation shrinks. Here the cluster shape is fixed first, its exact electrostatic moment is certified second, and a full-operator Maxwell estimate controls its subsequent homothetic size limit.

Mann and Alù's 2023 conference contribution on maximum asymmetric absorption in bianisotropic particles is an additional priority lead [13]. Only its official abstract and program information were available in this review. Its constitutive class, normalization, and relation to the present bounded dielectric size limit have not been fully checked.

No matching geometry-uniform quartic theorem was located in the inspected literature. This

absence is not a proof of novelty. The defensible result of this manuscript is the complete stated mathematical theorem and its finite-object realization. Any claim of scientific priority must await a more exhaustive literature and independent human review.

9 Adversarial audit, scope, and preserved research record

The uniform estimate was independently rederived from the dyadic kernel, coercive inverse, and quadratic radiation norm. The finite-sphere moment and interaction certificate were independently checked by two additional research agents, and the explicit Maxwell remainder constants were separately rederived. A mistaken draft description of the minimum enclosing radius was found and corrected; it was a length-convention error, not a change to the coefficient normalized by center separation. These are ordinary mathematical audits assisted by AI, not formal proof assistant certification or independent human peer review.

Potential failure	Resolution or explicit limit
Independent optimizing currents	Both sources use the same local χ and the same exact inverse. The extra factor occurs in (22).
Arbitrarily fine/disconnected geometry	Zero extension, Fourier projection, rearrangement, Schur bounds, and loss coercivity require only measurability and the stated caps.
Small or vanishing volume fraction	No lower volume fraction or extinction threshold is used. Equation (23) handles the normalization uniformly.
Missing radiation channels	Exact angular Fourier projection includes both polarizations and all electric-current multipoles. Numerical truncations are only checks.
Unjustified electrostatic approximation	The complete operator difference is bounded; the exact finite-sphere coefficient and a finite-t Maxwell remainder are controlled separately.
Illumination-gradient multipoles	The affine sphere moment identity applies to every L^2 input; the observation-phase cancellation occurs in the exact angular kernel.
Lossless responding sectors	Excluded by the full tensor floor $B \succeq mI$. A floor on only one region or only some polarizations is insufficient.
High-contrast or low-loss size sequences	The constants M, m must remain fixed. Letting $M \to \infty$ or $m \to 0$ can invalidate a uniform quartic conclusion.
Elliptical time-reversed polarization	Excluded. The order-zero electric drives can differ, so the same argument gives no extra factor. Earlier cubic examples remain separate.
Magnetic or bianisotropic constituents	Excluded. Their reversal changes part of the incident constitutive source at order one; a different theorem would be required.
Claiming every object is quartic	False: symmetric objects can have $C = 0$ and even $\Gamma = 0$. Sharp exponent is a statement about the admissible class.
Claiming an optimal coefficient	Not proved. The analytic coefficient interval is for one explicit object; the global finite-resource prefactor remains unresolved.

Potential failure	Resolution or explicit limit
Priority of sphere asymmetry	Already precedented in Sounas–Alù and heterodimer literature. No geometry or mechanism novelty is asserted.

Recovery and preservation. The recovery found ten named OQ7 deliverables: five earlier source archives, four PDF manuscripts, and the latest standalone fixed-resource LaTeX manuscript. Their bytes are preserved in the new archive's `originals/` directory with SHA-256 checksums. The most recent saved fixed-resource version was dated 10 October 2026 and contained the previous general-polarization cubic result. No saved quartic proof or two-sphere calculation was recovered. The present derivation and solver were therefore reconstructed and independently audited; the interrupted claim was not used as evidence for correctness. Earlier results, including unrelated planar and formalization work, were not edited or promoted into premises for the theorem here.

Reproduction. The archive includes the complete LaTeX source, the compiled manuscript, original Python source for Maxwell boundary matching and independent Mie/symmetry checks, exact-rational certificate checks, the complete numerical sweep and residuals, the figure script, and the audit notes. The included reproduction script uses Python with NumPy, SciPy, and Matplotlib; compiling the manuscript additionally needs a standard LaTeX distribution. Reproduction regenerates the checks and figures without editing the preserved originals.

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